



Cloud Computing

Cloud Computing: An Introduction

- Cloud computing evolved from earlier forms of distributed computing.
 - Clusters, grids, and the notion of resource sharing over a network are all forerunners of clouds.
- Clouds support High Performance Computing (HPC) and High Throughput Computing (HTC)
 - **HPC:** use of parallel processing to execute large programs quickly; often equated to supercomputers, typically applied to systems generating teraflops (10^{12}) or more. High speed computing.
 - **HTC:** running a job that takes days to complete, or an application that must produce a high number of completed operations per unit of time.

What is Cloud Computing?

- “...a style of computing in which dynamically scalable and often virtualized resources are provided as a service over the Internet.”
- Clouds provide a level of abstraction that simplifies user access to the services it offers.

What is Cloud Computing?

Cloud computing covers both services provided and the software and hardware in the datacenters that provide the services.

“From a hardware perspective three aspects are new in Cloud Computing.

1. The illusion of infinite computing resources available on demand, thereby eliminating the need for Cloud Computing users to plan far ahead for provisioning.
2. The elimination of an up-front commitment by Cloud users, thereby allowing companies to start small and increase hardware resources only when there is an increase in their needs.
3. The ability to pay for use of computing resources on a short-term basis as needed (e.g., processors by the hour and storage by the day) ...” [“Above the Clouds: a Berkeley View of Cloud Computing” , <http://www.eecs.berkeley.edu/Pubs/TechRpts/2009/EECS-2009-28.pdf>]

What is Cloud Computing?

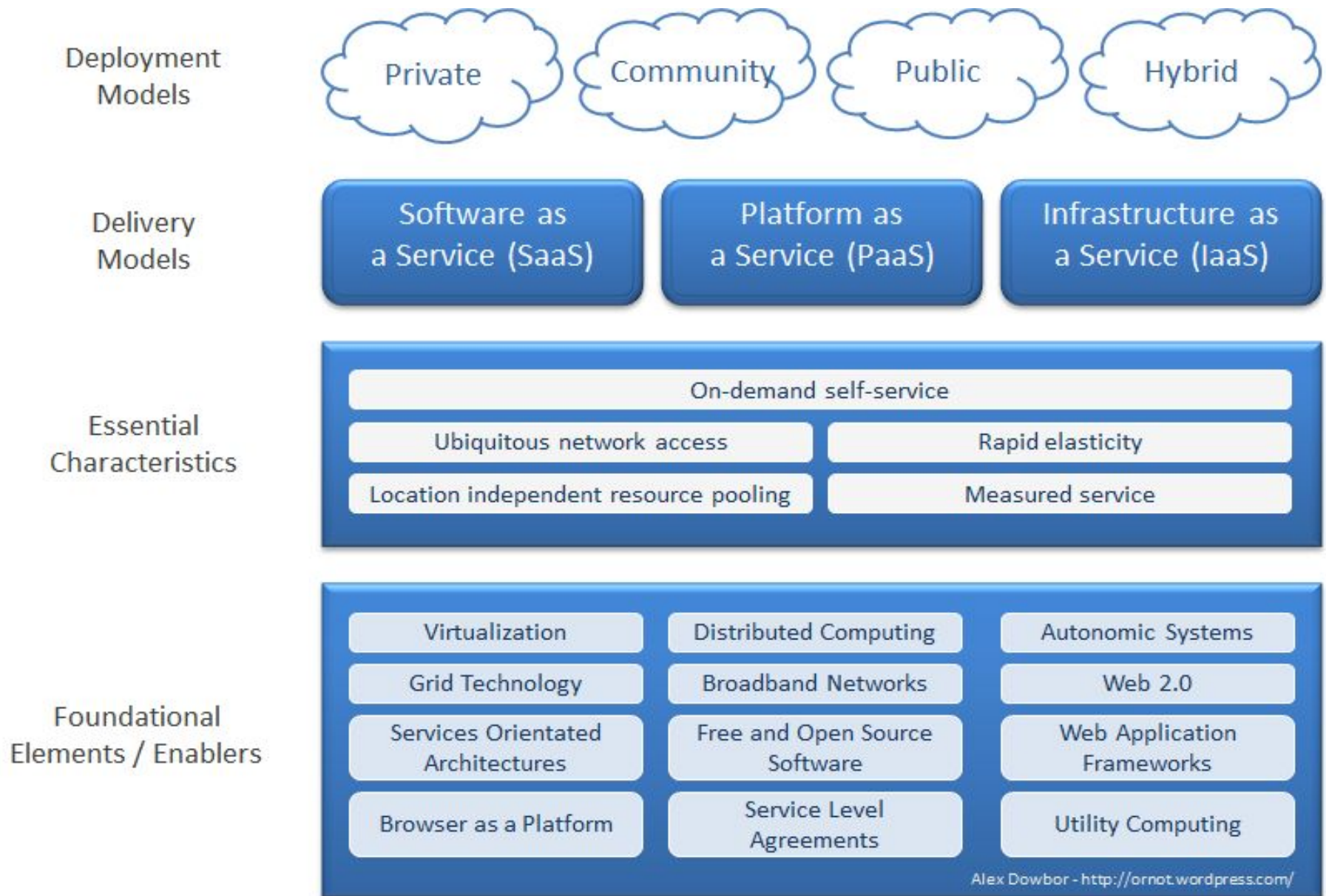
- The cloud consists of datacenter hardware, system software, and a variety of services.
- The datacenters can be centralized or distributed.
- NIST, the National Institute of Standards and Technology, prepared the following working definition of cloud computing



NIST Definition

- “... a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction.”
- The NIST definition also identifies
 - 5 essential characteristics
 - 3 service models
 - 4 deployment models





NIST Working Definition of Cloud Computing

- Deployment Models: how are cloud resources arranged, managed
- Delivery Models
- Essential Characteristics
- Enabling Technologies



Cloud Deployment Models

from NIST, the National Institute of Standards and technology

- **Public:** Accessible, via the Internet, to anyone who pays
 - Owned by service providers; e.g., Google App Engine, Amazon Web Services, Force.com.
- **Community:** Shared by two or more organizations with joint interests, such as colleges within a university
- **Private:** Accessible via an intranet to the members of the owning organization
 - Example: NASA's cloud for climate modeling
<http://www.networkworld.com/news/2010/020110-nasa-cloud-service.html>
- **Hybrid**
 - A private cloud might buy computing resources from a public cloud.
- Currently, public clouds are the most common although over time this may not be the case. Private clouds are easier to manage and provide greater security.

NIST Working Definition of Cloud Computing

- Deployment Models
- Delivery Models: how are services offered to cloud users?
- Essential Characteristics
- Enabling Technologies



Delivery Models: IaaS

- Users interact with clouds in various ways to access its various services.
 - Subscription-based, pay-as-you-go
 - Rely on SLAs (Service Level Agreements) that specify guaranteed levels of service, bandwidth, security, etc.
- Service Delivery Models
 - IaaS: Infrastructure as a service
 - PaaS: Platform as a service
 - SaaS: Software as a service

Delivery Models: IaaS

- Provides access to (possibly virtualized) resources: servers, storage devices, networks..
 - Includes *storage as a service*, *compute instances as a service*, *communication as a service*.
- The customer can deploy his own applications, databases, operating system environments, and so on
- The customer does not necessarily have control over the underlying hardware

IaaS Benefits

- Organizations don't need to maintain their own data centers
 - Saves on staffing, reduced hardware costs, etc.
- Resources can be scaled up (or down) easily
- Possible security benefits
 - Private data centers = single point of failure
 - Use a cloud data center as a less-costly backup to the on-site data center

IaaS Example

- Amazon EC2 (Elastic Compute Cloud)
<http://aws.amazon.com/ec2/>
 - Users can “rent” virtual machines (called *instances*) and use them to run any software
 - Instance types: on-demand (pay as you go, no commitment), reserved (pay in advance for a certain amount of time), spot (be the lowest bidder for unused time)
 - “Elastic” refers to the ability to scale from 1 instance to hundreds in a matter of minutes.

Delivery Models: PaaS

- In addition to infrastructure, PaaS supplies operating systems, programming languages, and other services which the customer can use to create and deploy applications.
- As before, the user doesn't directly control the hardware or system software, but can control his own applications.

PaaS Benefits

- In addition to the benefits of IaaS (scalability, economy of scale, lower overhead, etc.) PaaS provides the user with current system software. Updates, patches, other issues are handled automatically.
- IT department can focus on application development and do not need to deal with server-related issues.

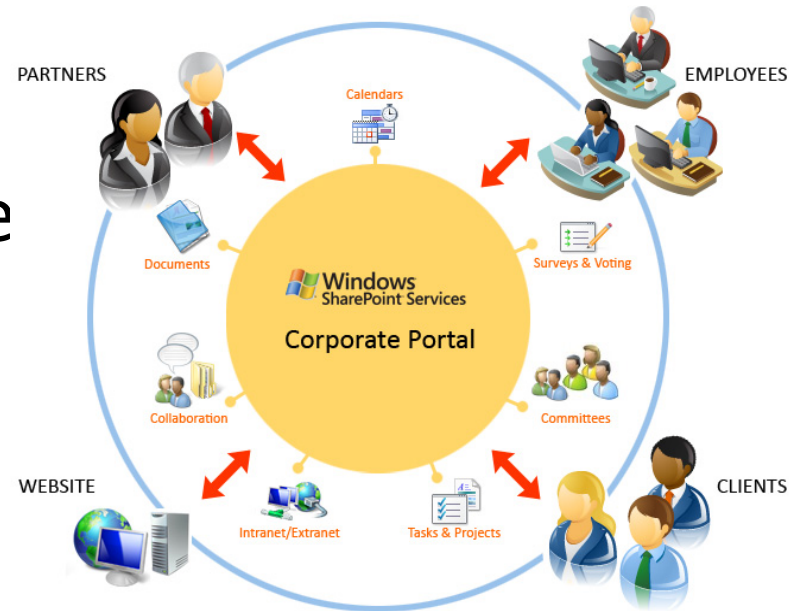
PaaS Example

- Force.com is a scalable and secure cloud platform from Salesforce.com, a company that specializes in customer relation software (CRM) tools.
<http://www.force.com/> , <http://en.wikipedia.org/wiki/Force.com>
- It delivers a complete technology stack: from database and security to workflow and user interface.
- Simplifies the development of enterprise applications:
 - “A comprehensive array of technologies built on top of the Force.com API dramatically simplifies enterprise application integration. Whether you need to connect to SAP, Oracle, Microsoft, or another on-premises or cloud-based application, integration now takes much less effort than ever before”

Delivery Models: SaaS



- Complete software applications are offered
 - Customers access apps directly through Web interfaces.
 - Example: Gmail, SharePoint
- SaaS builds on the software and resources in the Paas and IaaS layers.



Delivery Models: SaaS

- SaaS is similar to, but more advanced than, the Application Service Providers (ASP) that host and manage software applications for businesses.
 - ASPs managed 3rd party software, SaaS providers develop their own
 - ASP generally provided a separate instance of the software/hardware platform for each customer but SaaS often supports multitenant solutions.

SaaS: Multitenant Architecture

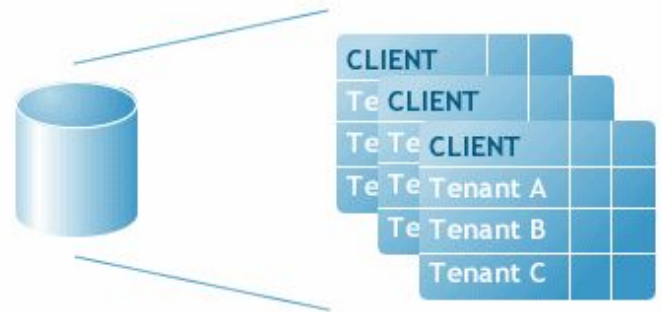
- Customers may share the same server & even the same software; e.g., a database
 - Data is partitioned and remains private
- Saves money; e.g., only pay for one license, only maintain one copy, ...
- Compare to virtualization, where each customer *appears* to be running on a single machine.

Figure 1

SINGLE TENANT
Separate Databases



MULTI-TENANT
Shared Databases,
Identical Schemas



MULTI-TENANT
Shared Databases,
Custom Schemas

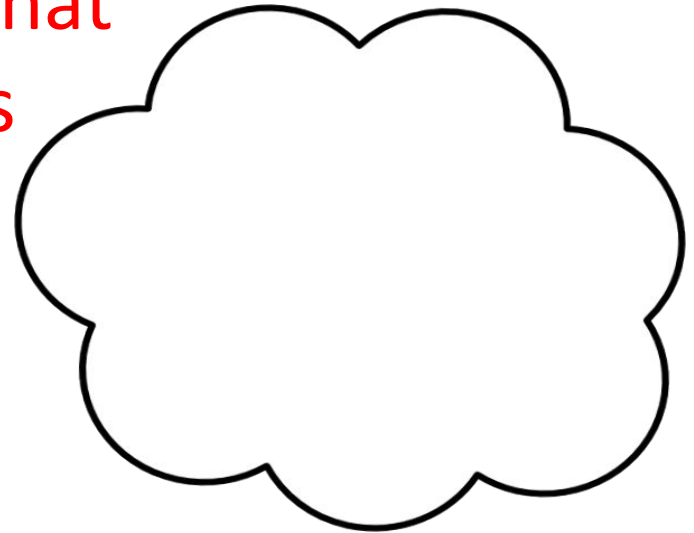


SaaS: Advantages

- Reduce or eliminate need for on-site data center
- Pay-as-you-go model, no big up-front investment in either hardware or software
- Scalability in number of servers, amount of storage, etc.
- Device-independent access to applications
- Improved disaster recovery, less interruption, (ideally – but ... what about Google outages?)

NIST Working Definition of Cloud Computing

- Deployment Models
- Delivery Models
- Essential Characteristics: what features should cloud users expect?
- Enabling Technologies



Characteristics of Clouds

(NIST report)

- On-demand self-service: consumers can acquire the necessary computational resources without having to interact with human service providers.
- Ubiquitous network access: cloud features don't require special devices – laptops, mobile phones, etc. are generally supported.
- Resource pooling: cloud resources are pooled to serve many customers “... using a multi-tenant model, with different physical and virtual resources...”

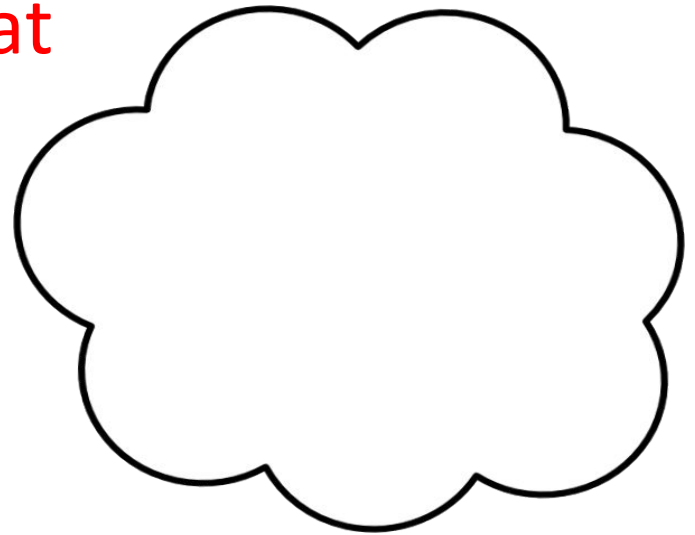
Characteristics of Clouds

(NIST report)

- Rapid elasticity: resources can be allocated and de-allocated quickly as needed.
- Measured service: resource use is measured and monitored; charges are made based on usage and service type (e.g., storage, CPU cycles, etc.)

NIST Working Definition of Cloud Computing

- Deployment Models
- Delivery Models
- Essential Characteristics
- Enabling Technologies: what are the underlying capabilities of the cloud?



Enabling Technologies & Concepts

- Virtualization

- Supports virtual clusters, sharing of hardware among several users, load balancing through VM migration, etc.
- Multiple servers can share a single computer or a shared cluster for improved CPU utilization.
- Support for legacy systems based on old operating systems and applications
- Customers can choose their preferred OS; different apps with different OS's can run on the same hardware.
- Example: Amazon's EC2 provides an API that allows users to configure virtualized hardware to support any kind of application.

Enabling Technologies & Concepts

- Clustering/grids

- “...enables scalable parallel and distributed computing...”
- Clouds make use of existing data centers, based on cluster and grid technology
- Combining existing hardware with virtualizing software makes it possible to provide the elasticity that characterizes cloud computing.
- Large data centers



[SGI Ice Cube data center](#)

Outside: 20' x 40', inside: an up-to-date data center complete with power, cooling facilities, racks for SGI servers and storage systems, et.

Enabling Technologies & Concepts

- Autonomic systems: self-managing; the ability of a system to make decisions based on built-in policies.
 - Self-monitoring, self-correcting
 - Hides the complexity of distributed systems
- Example: load balancing
 - Demands for a given service vary over time
 - During slack periods one virtual server may be sufficient; when client requests increase an autonomic system can allocate a load balancing server to direct requests to two or more servers in a round-robin fashion. When demand decreases, return to original level.

Enabling Technologies & Concepts

- Service Oriented Architectures (SOA)
 - A software architecture describes the components of a software system and connections between components. It deals with external, not internal issues, e.g. APIs.
 - In SOAs, the components include various *web services*, self-contained, modular components that communicate in some standardized way (XML, SOAP, ...) and can be easily integrated into other applications.
 - The World Wide Web Consortium defines a web service as “a software system designed to support interoperable machine-to-machine interaction over a network ”
 - Web services interface to each other, not users. A developer builds an application from web services and provides it with a GUI to interface to users.

Enabling Technologies & Concepts

- Web Services - Restful
 - RESTful web services (Representational State Transfer): (Hwang) expose a set of resources (documents, templates, etc.) identified by a name (URI). HTTP is the standard interaction protocol. Interactions are stateless. Used in Web 2.
 - More heavy-duty Web services are build on SOAP, WSDL, and UDDI

Virtual Machine, Guest Operating System, and VMM (Virtual Machine Monitor) :

Virtual Machine

A representation of a real machine using software that provides an operating environment which can run or host a guest operating system.

Guest Operating System

An operating system running in a virtual machine environment that would otherwise run directly on a separate physical system.

The Virtualization layer is the middleware between the underlying hardware and virtual machines represented in the system, also known as *virtual machine monitor* (VMM).

VMM : Virtual Machine Monitor

Virtual Machine Monitor

Software that runs in a layer between a hypervisor or host operating system and one or more virtual machines that provides the virtual machine abstraction to the guest operating systems. With full virtualization, the virtual machine monitor exports a virtual machine abstraction identical to a physical machine, so that standard operating systems (e.g., Windows 2000, Windows Server 2003, Linux, etc.) can run just as they would on physical hardware.

Example:

Stand alone controllers that monitor building HVAC, electrical, and other systems can be networked to provide integrated building automation and energy management.

“Smart” buildings

e.g., reduce power consumption by adjusting lighting & heat based on current occupancy

Enabling Technologies & Concepts

- Utility Computing: the business model in which consumers pay for services according to the amount and type used.
 - “Plug into the cloud” and connect to your service/platform/hardware
 - The service provider monitors usage and bills on a monthly or other basis.
- Grids and clouds are both classified as utilities, but clouds offer more features than the typical grid, and usually support easier interaction.
- Utility computing is a feature of public clouds.



Enabling Technologies & Concepts

- Browser as a Platform
 - A computing platform includes a hardware architecture and software framework (including applications), where the combination allows software, particularly application software, to run.
 - “Browser as a platform” indicates the use of a browser as the interface to application software.
- Other technologies: broadband networks, free & open source software, service level agreements, etc.

Cloud Design Objectives

- Data storage, processing, software moves off the desktop and into the cloud
- Users enter into Service Level Agreements (SLAs) with providers and pay based on the SLA
- Cloud must be able to scale resources and services as the user's needs grow (or shrink)
- Privacy must be guaranteed!
- Quality of service should be standardized.
- Standards must be agreed on to prevent *data lock-in*, which means that once a customer is committed to one service provider it is not possible to shift to another.

Other Cloud Topics

- In the technical report “Above the Clouds: ...” Armbrust et al. address a number of issues related to cloud computing.
- Brief overview of a few:
- The economics of cloud computing
- Obstacles and opportunities
- Clouds as a facilitator of new services

The Economics of Cloud Computing

- Cloud computing shifts capital expenses (for hardware, software to run large datacenters) to operating expenses for cloud services
- Potential to save money: an estimated server utilization rate of 5% to 20% is reported by Armbrust, et al. for private data centers primarily because users provision for peak loads.

Cloud Economics – Elasticity Example

- Assumptions:
 - A service requiring 500 servers at noon, 100 servers at midnight provisions for peak load (500)
 - Average utilization over 24 hours = 300 servers
 - Daily usage: $24 \times 300 = 7200$ server-hours
 - Actual capacity: $24 \times 500 = 12000$ server-hours
 - Capacity/usage ration: 1.7
- Compare the pay-as-you-go cost of utility computing for 3 years (equipment lifetime) to the cost of buying the servers, go with the cloud if it will save money.

Cloud Economics – Transferring Risk

- Suppose a private data center is sized to meet the expected peak load, but some event (e.g., exciting new product, great reviews) drives even more customers to the site.
- Of the potential new customers turned away because of overloading, some percentage will never return
- Loss of potential customers plus possible bad press could have been averted with a more flexible (i.e., cloud) system.
- For more about economics, read the Armbrust report.

Obstacles and Opportunities

- Service availability:
 - consumers fear the loss of availability either through data center failure or simple under-provisioning of the cloud
 - Distributed denial of service attacks by criminals who rent bots to launch attacks & then demand payment to stop. See the Berkeley paper “Above the Clouds...” for 9 more
- Defense
 - Use more than one company
 - Scale up to a larger number of computers which will cost the attackers more, making it difficult to sustain the attack.



Data Centers

A centralized collection of data, usually related to a single business or subject
for example: Google data centers, National Climate Data Center

Data centers typically provide data storage and management and make the data available to the public, a paying client, ...

Data centers in cubes are mobile



A Container Cube Data Center

Built-in climate control, power, etc. makes container cubes self-sufficient.

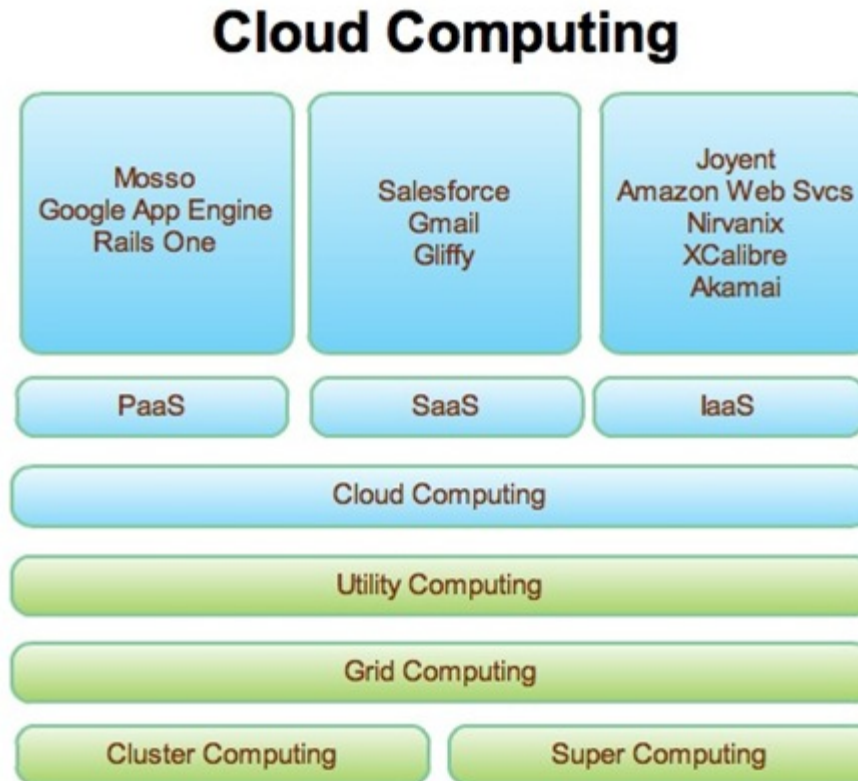
No need for a building

Mobility – able to move to a more congenial environment.



Courtesy: Dan Reed, Microsoft

Another Look at Cloud Technology



1.0 In blue you have what is lately called Cloud Computing. In green, some of the underlying work done that led to Cloud Computing. At the top are examples of each XaaS type.